MIL-P- 18140 (SHIPS)
29 November 1954
SUPERSEDING
S47-3 (in part)
(See 6.3)

MILITARY SPECIFICATION

PUMPS, ROTARY, FUEL OIL TRANSFER, POWER DRIVEN

1. SCOPE

1.1 Scope. - This specification covers positive displacement rotary fuel oil transfer pumps, class N-1, including prime movers for.

1.2 Classification. - Class N-1 positive displacement rotary pumps shall be of the f ollowing types:

Type I - Cam and plunger.
Type II - Screws with timing gears.
Type III - Screws without timing gears.
Type IV - Impellers with timing gears,

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

FF-B- 171- Bearing, Ball, Annular, (General Purpose).
GGG-P- 15- Packing Tools, Extracting and Tamping; Stuffing Box Packing and Lantern Glands.
GGG-P-781 - Pullers, Bushing, Bearing, Gear, Wheel Hub and Cylinder Sleeve, Installing and Removing.

MILITARY

MIL-B-233 - Boxes, Repair Parts.
JAN-P-735 - Primer, Paint, Zinc-Chromate, Alkyd Type.
MIL-S-869 - Steel: Forgings for Nitriding and Nitrided Steel.
MIL-S-890 - Steel: Forgings and Bars for Hulls, Engines, and Ordnance (Heat Treated).
MIL-N-894 - Nickel-Copper-Alloy Bars, Plates, and Other Wrought Forms.
MIL-S- 901- Shockproof Equipment, Class HI (High -Impact) Shipboard Application, Tests for.
MIL-D- 963 - Drawings, Production (for Electrical and Mechanical Equipment for Naval Shipboard Use).
MIL-C-2174 - Controllers and Master Switches, Direct-Current (Naval Shipboard Use).
MIL-C-2212 - Controllers and Master Switches, Alternating- Current (Naval Shipboard Use).
MIL-I-15024 - Identification Plates, Information Plates and Marking Information for Identification of Electrical, Electronic and Mechanical Equipment.
MIL- T- 15071 - Terminal Manuals for Mechanical and Electrical Equipment.

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MILITARY (cont'd.)

MIL-S-15083 - Steel; Castings.
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 111).
MIL-R- 15137 - Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-E-15163 - Engines, Diesel, Commercial-Type.
MIL-C-15328 - Coating, Pretreatment (Formula No. 117 for Metals).
MIL-B-16261 - Bronze, Bearing: Castings.
MIL-13-16540 - Bronze, Phosphor: Castings,
MIL-M-16576 - Metal, Gun: Castings,
MIL-P-16789 - Preservation, Packaging, Packing and Marking of Pumps, General, and Associated Repair Parts,
MIL-M-17060 - Motors; Alternating-Current, Integral HP (Shipboard Use).
MIL-I- 17166- Iron Castings; Nodular Graphitic (Ductile Iron) (for Shipboard Applications).
MIL-M-17413 - Motor, Direct-Current, Integral HP (Shipboard Use).
MIL-N-17506 - Nickel- Copper-Aluminum Alloy, Wrought.
MIL-T-17523 - Turbine, Steam, General Auxiliary (Naval Shipboard Use).
MIL-P-17545 - Primer, Paint, Shipboard, Alkyd-Red Lead Type, Formula No. 116.
MIL-G-17859 - Gear Assembly, Speed Decreaser; and Gear Assembly, Reverse (Naval Shipboard Use).
MIL-G-18121 - Gear Assemblies, Speed Decreaser, Electric Motor Driven Naval Shipboard Auxiliary Machine Application.

NAVY DEPARTMENT

General Specifications for Inspection of Material.
General Specifications for Machinery:
Section S1-4 - Welding and Allied Processes.
42B10 - Bearings, Roller.
46B27 - Copper-Silicon-Alloy, Wrought: Bars, Forgings, Plates, Rods, Shapes, Sheets and Strips.
46-I-5 - Iron, Gray, High-Test: Castings.

STANDARDS

MILITARY

MIL-STD-10 - Surface, Roughness, Waviness and Lay.

DRAWINGS

BUREAU OF SHIPS

B-104 - Flanged Composition Pipe Fittings.
B-105 - Flanged Composition Pipe Fittings.
B-153 - Packings and Gaskets, Application of.
B-176 - Flanges, Composition, Silver Brazing.
B-177 - Flanges, Composition, Silver Brazing.
B-214 - Root Connections for Attaching Piping.
5000-S4800-3000 - Piping Schedule-Pipe Fittings, Valves and Types of Joints Used in Piping Systems.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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2.2 Other publications. - The following document forms a part of this specification, Unless otherwise indicated the issue in effect on date of invitation for bids shall apply,

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C,)

3. REQUIREMENTS

3.1 Qualification. - The power driven pumps furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein, (See 6.2.)

3.2 Material. -

3.2.1 Material shall be as specified herein, Material not definitely specified shall be of the best quality used for the purpose in commercial practice, and shall be subjected *to* review and approval by the bureau or agency concerned. Material shall be free from all defects and imperfections that might affect the serviceability of the finished product,

3.2.2 The use of materials other than those indicated will be considered when the contractor shows the necessity for the substitution and the material proposed is readily obtainable, Substitution of materials will require the specific approval of the bureau or agency concerned in each case.

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3.2.3 Materials for the pumps shall be in accordance with table I, as applicable.

Table I - Materials for class N-1 pumps.

Application	Material	Specification
Casing and heads	Cast steel, class B	MIL-S-15083
	Gun metal	MIL-M-16576
	Nodular iron	MIL-I-17166
Shafts	Forged steel, alloy No. 1 or No. 2	MIL-S-890
	Nitrided steel, class a	MIL-S-869
	Nickel-copper-aluminum alloy	MIL-N-17506
	Nickel-copper-alloy, rolled, class a or b	MIL-N-894
Rotors	Forged steel, alloy No. 1 or No. 2	MIL-S-890
	Nitrided steel, class a or b	MIL-S-869
	High test grey iron	46-I-5
	Nodular iron	MIL-I-17166
	Gun metal	MIL-M-16576
	Copper-silicon-alloy	46B27
Sleeve bearings	Bearing bronze, grade II	MIL-B-16261
	Phosphor bronze, grade A	MIL-B-16540
Rotor housings or cylinder liners	Bearing bronze, grade II	MIL-B-16261
	High test grey iron	46-I-5
	Nodular iron	MIL-I-17166
Timing gears	Forged steel, alloy No. 2	MIL-S-890
	Nitrided steel, class a or b	MIL-S-869
Casing side wearing plates	Chromium cobalt placed, 300 Brinell hardness minimum	
	Nitrided steel	MIL-S-869
Stuffing-box glands	Phosphor bronze, grade A	MIL-B-16540
	Bearing bronze, grade I	MIL-B-16261
	Gun metal	MIL-M-16576
Piston slide pins	High test grey iron	46-I-5
	Nodular iron	MIL-I-17166

3.3 General design. - The design and construction of the pumps shall be the lightest and most compact possible consistent with the requirements of reliability, accessibility to repair, resistance to corrosion and erosion, and economy. The principle of reliability, however, is paramount and no compromise of this principle shall be made with the other basic requirements of design. The pumps shall be of such design that they will operate over a long period of years with a minimum of wear or erosion of any operating parts, such as casings, liners, end plates, rotors, gears, shafts, glands and bearings, The parts subject to unavoidable wear or erosion shall be of the best material available for the purpose seas to reduce these detrimental effects to a minimum.

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3.3.1 The pumps shall operate in a manner satisfactory to the bureau or agency concerned when inclined as follows:

(a) Up to 15 degrees to either side (permanently inclined).
(b) With the vessel rolling up to 45 degrees from the vertical to either side
(c) With the vessel pitching 10 degrees up and down from the normal horizontal plane.

3.3.2 Shockproof ness. -

3,3.2.1 All equipment shall be designed to withstand shock due to firing of the ship's own armament and non'- contact underwater explosions of near-miss aerial bombs, torpedoes, and mines. Every attempt shall. be made to design the equipment to resist shock as specified by graphs, A, B, and C, shown on figure 1, it being understood that these shocks are applied at the footings of the component, Shockmounts shall not be used unless bureau or agency concerned approval of the mount and its proposed application is given.

3.3.2.2 Equipment shall be designed to resist shock by using the accelerations specified on the graph shown on figure 1 for determining shock forces. The determination of stresses resulting from these forces is accomplished by the application of the usual methods of analytical mechanics. The allowable stress in components due to shock shall be the yield strength (O. 2 percent offset) of the material. Normal stresses in parts, such as the tensile stress in a bolt due to initial pull-up, shall not be added to the shock stresses, It *is* of paramount importance that the accelerations shown on the curves of figure 1 be utilized in the design of holding down bolts, supporting feet or lugs and main structural members of the equipment and that no failure or permanent deformation occur as a result of shock, Components of an assembly mounted on a common rigid base need not be designed for higher accelerations than the values for which the whole assembly is designed unless the designer has reason to believe that higher accelerations will be experienced by the components.

3.3.2.3 The design of all complete pump units shall be such that they are capable of passing the high-impact shock test specified in Specification MIL-S-901.

3.3.2,4 Bolts designed to be stressed in shear shall be installed in holes with a minimum of clearance, as experience indicates that large clearances allow impacting and subsequent failure.

3.3.3 Alignment between separate components of a unit shall, in general, be maintained by means of keyways or other adequate means.

3.3.4 In no case shall a piece of equipment be rigidly supported from more than one plane.

3.3.5 The design, construction, and workmanship, of the equipment shall be such that a minimum practical noise level will result under all operating conditions.

3.3.6 The pumps shall be self-priming and capable of removing air from the suction lines, The pumps may at times be required to handle oils lighter than those specified, water, sediment and sludge, such as may settle out and accumulate in the bottom of tanks: Hence, where the lubrication of the pump is supplied by the fluid pumped, the bearings, rotors, and gears shall be of such design and materials that the pumps will operate in a manner satisfactory to the Bureau of Ships when pumping such fluids for limited periods (see 4. 2).

3.3.7 Mounting. -

3.3.7.1 Pump units shall be horizontally or vertically mounted, as specified (see 6. 1).

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3.3.7.2 Each horizontal pump and driving unit shall be mounted on a common bedplate.

3.3.7.3 Each vertical unit shall be base mounted or table mounted, as specified (see 6. 1).

3.3.7.4 For the base mounted units, the base may be either integral with the pump casing, or a separate piece, and the base shall have provision for bolting to foundation,

3.3.7.5 For the table mounted units, the table mount may be either integral with the pump casing or bracket, or it may be a separate piece.

3.3.7.6 A suitable bracket shall be provided between the pump and driver of vertical units.

3.3.7.7 Bedplates, bases, table mounts and brackets shall be of cast steel, class CW of Specification MIL-S-15083, or structural steel fabricated by welding, designed so that they shall be sufficiently rigid to permit handling, shipment and installation of the unit onboard ship, without disturbing the-alignment of the assembled units, and such that the normal distortion, weaving or vibration of the supporting structures onboard ship cannot cause misalignment between the pumps and driving units.

3.3.7.8 Each component part of an assembled unit supported directly by the bedplate, base, table mount, or bracket, shall be doweled thereto to facilitate reassembly and maintenance of alignment.

3.3.7.9 All bearing and seating surfaces of bedplates, bases, table mounts, or brackets shall be finished machined.

3.3,7.10 In the event special bulkhead mounting is required, sideplates shall be furnished subject to all the conditions specified for basepla tes.

3.3.7.11 Horizontal pumps shall be designed so that all bearings, rotors, and shafts may be removed and replaced without disconnecting piping, except stuffing box gland seal piping, and without removing the pump or its driving unit from the bedplate or base.

3.3.7.12 Each vertical unit, in order to facilitate interchangeability, shall be constructed to permit rotation of the pump with respect to its driver. Eight or more positions of pump flanges relative to the driver shall be available.

3.3.8 Bearings.-

3.3.8.1 Radial bearings. - Radial bearings may be of the rolling contact or sliding surface (sleeve) type.

3.3.8.2 Rolling contact bearings for radial loads and the installation thereof shall be in accordance with Specification FF-B- 171 and 42B10.

3.3.8.3 Sliding surface radial bearings may be of the fixed or self aligning type as approved for design and application by the bureau or agency concerned. Ample bearing surface shall be provided for each sleeve bearing.

3.3.8.4 Thrust bearings, - Pumps shall be designed to counteract any unbalanced hydraulic or mechanical thrust under normal service condition, In this connection consideration shall be given to the fact that the normal motion of a ship at sea or an abnormal list or trim giving an inclined operating platform may introduce mechanical thrust in a unit which might otherwise be in satisfactory balance.

3.3.8.5 Thrust bearings may be of the straight thrust rolling contact, combined radial and thrust rolling contact or plain sliding surface type.

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3.3.8.6 Thrust bearings of the rolling cantact type and the installation thereof shall be in accordance with Specifications FF-B-171 and 42B10, as applicable.

3.3.8.7 Sliding surface thrust bearings shall be as approved for design and application by the bureau or agency concerned,

3.3.8.8 Bearing installation. -

3.3.8.8.1 Where sliding surface (sleeve) bearings are fitted in closed heads they maybe of the solid bushing type secured by pins or keys, For independent bearings on brackets fitted at stuffing box ends, the sliding surface bearings shall be split along the axis, All sliding surface bearings shall be so arranged to permit renewal or refitting without removal of the pump rotor from the casing, and they shall be securely fitted to prevent axial movement.

3.3.8.8.2 For either internally or externally mounted bearings, the bearing housing shall be cast integral with the pump case or bearing bracket, or secured thereto in such a manner as to insure alignment.

3.3.8.8.3 Bearing brackets shall be either cast integral with the casing, securely bolted to the casing on a machined shoulder, or held in alignment with the casing by at least two heavy dowels and securely bolted thereto. The use of bolts alone for securing brackets is prohibited.

3.3.8.9 Bearing lubrication. -

3.3.8, 9.1 Rolling contact or sliding surface bearings may be lubricated by the fluid pumped where such bearings are located within the pump casing and are not separated by packed glands from the fluid pumped, provided the bearing housings are so designed that there are no pockets, which may act as water, sediment, or sludge traps and a continuous circulation of fresh oil is provided through such bearings.

3.3.8.9.1. 1 Bearings lubricated by the fluid pumped shall be required to operate in a manner satisfactory to the Bureau of Ships throughout the test specified in 4.2.7. 1; however deterioration to a point requiring replacement of the bearings immediately after the test will not disqualify the bearings,

3.3.8.9.2 Externally mounted rolling contact bearings shall be lubricated in accordance with Specification FF-B-171 or 42B10, as applicable.

3.3.8.10 The number, type and arrangement of bearings shall be as approved by the bureau or agency concerned,

3.3.9 Piping and valves. -

3.3.9.1 All valves, flanges and fittings for pipe connections shall conform to Drawing 50 CO-S4800-3000.

3,3. 9.2 All suction, discharge, and relief valve connections shall be flanged; the flanged connections shall be in accordance with Drawings B-104, B-105, B-176, and B-177, as applicable, Suction and discharge pressure gage connections shall be flanged; the flange connections shall be in accordance with Drawing B-214.

3.3.9.3 Gland seal piping shall be 1/8 inch iron pipe size (i. p. s.) extra heavy steel pipe, steel valves, steel fittings and steel unions, all screwed.

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3.3.10 Drains. - The bottom of all pump cacings or head shall be provided with a drain connection. Drains shall be fitted to drip pockets,

3.3.11 Casings and heads. -

3.3.11.1 Casings shall be made as light as possible consistent with strength and shall be ribbed and braced on the outside as necessary to give stiffness and rigidity under the working pressure. The interiors shall be smooth cored with easy bends wherever the direction of flow is changed,

3.3.11.2 It is preferred that pump casings be equipped with easily removable liners, However, consideration will be given to pumps not having this feature. Either liners or cylinders shall be supplied as repair parts as specified in 3.9 covering wearing parts.

3.3.11. 3 Suction and discharge openings snail be cast integral with the casings or heads, shall have machined and scraped or ground flange faces suitable for metal to metal joints, and shall be located as approved by the bureau or agency concerned,

3.3.11.4 The heads shall have accurately machined faces with a ground or scraped finish. Joints between casings and heads may be made up either metal to metal or with a thin manila paper gasket not exceeding 0.007 inch in thickness, Heads shall be rabbetted or doweled to the casings.

3.3.11.5 Forcing bolts and lifting bolts shall be provided as necessary for convenient handling during overhaul.

3. 3.11.6 One or more test lugs may be required to be cast integral with gun metal castings in accordance with Specification MIL-M- 16576, at the discretion of the bureau or agency concerned. When test lugs are required the bureau or agency concerned will advise in its comments on working drawings as to the number, size and location of the test lugs required.

3.3.12 Rotors and shafts. -

3.3.12.1 All rotors shall be accurately finish machined or ground throughout.. They shall be secured to the shafts by keys or other method approved by the bureau or agency concerned, Rotors shall be secured positively against lateral displacement on shafts.

3.3.12.2 Means shall be provided for the escape of the liquid pumped from all points between rotors or gear teeth where it may be trapped during operation and thereby create a dangerously high pressure. Means of entrapment relief shall be satisfactory to the bureau or agency concerned.

3.3.12.3 Deflector rings or other similar method approved by the bureau or agency concerned shall be used to prevent leakage from the stuffing boxes running along the shafts. The use of felt seal rings only will not be approved; if authorized in combination with other methods, they shall be installed so they can be replaced without dismantling the pump,

3.3.12.4 All shafts shall be accurately machined and ground. Means shall be provided on the driven or driving shaft of each unit so that a portable tachometer may be quickly and easily applied or unless a revolution counter is permanently installed. Either the tachometer or revolutions counter shall by of a type which is satisfactory to the bureau or agency concerned, Provision for the use of tachometer will not be required on pumps driven by alternating current motors. Shafts shall have a minimum hardness of 275 Brinell.

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3.3.13 Timing gears,- Timing gears may be of the spur or single helical type. Timing gears of steel shall be case hardened to a minimum depth of 0. 0625 inch and shall have a hardness of 575 to 700 Brinell. Gears shall be fully enclosed and adequately lubricated. Timing gears shall be secured to the shafts by keys or other method approved by the bureau or agency concerned. .

3.3.14 Couplings. -

3. 3.14.1 Unless otherwise specifically approved by the bureau or agency concerned, a flexible coupling of all metal construction and of a design satisfactory to the bureau or agency concerned shall be installed between the pump and the driving unit,

3.3.14.2 All coupling hubs, which are to be f itted to driving or driven shafts larger than 1 inch in diameter, shall be fitted to the shafts on a taper, shall be secured against turning on the shafts by keys fitted parallel to the taper, and shall be secured on the shafts by loc k nuts.

3.3.15 Stuffing boxes and packing. -

3.3.15.1 Stuffing boxes shall be of adequate depth and so designed that when properly packed leakage is reduced to a minimum under all operating conditions.

3.3.15.2 Pump gland packing shall conform to the applicable specifications listed on Drawing B- 153. Sufficient space shall be provided between the bearings and stuffing boxes to permit ready renewal of packing and inspection and overhaul of the bearings.

3.3.15.3 Where a stuffing box is subject to discharge pressure, the design of the box shall be such as to trap the leakage therefrom by the use of a sealring, or other method approved by the bureau or agency concerned, and return same by drain piping to the suction side of the pump, with needle valve to control the flow where necessary. Where a stuffing box is subject to suction pressure, suitable means shall be provided by seal rings and seal piping connected to the discharge of the pump for sealing the packing to prevent leakage of air into the pump suction. The sealing piping shall be fitted with a needle valve or valves as necessary for proper control,

3.3.15.4 Glands and seal rings, if used, shall be split in halves with the halves secured together in a manner to form solid rings for setting up.

3.3.5.5 All stuffing box glands shall be set up by nuts threaded on gland swing bolts or studs secured in the casings.

3.3.15.6 Seal rings shall be constructed, insofar as practical, so that lantern gland extracting tools in accordance with Specification GGG-P-15 may be used.

3.3.16 Rubbing speeds, - The maximum rubbing speed in feet per minute for a particular type pump shall be the speed established for the type pump during the qualification test, Pumps shall not be offered which operate at a rubbing speed greater than the rubbing speed at which the pump of the same type operated on qualification test.

3.3.17 Threaded parts. - Threaded parts shall conform to Handbook H28,

3.3.18 Welding. - Welding shall be in accordance with Section S1-4 of the General Specifications for Machinery.

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3.3.19 Painting.- For pumps operating at a temperature of 300" F. or less, all external unmachined surfaces of ungalvanized ferrous metal parts shall be thoroughly cleaned and coated with one coat of pretreatment conforming to Specification MIL-C- 15328, and one coat of zinc chromate primer, Specification JAN-P-735, or one coat of red lead primer conforming to Specification MIL-P- 17545, followed by a finish coat of light gray equipment enamel conforming to Specification MIL-E-15090.

3, 3. 19. 1 painting of external surfaces of nonferrous parts of pumps will not be required.

3.4 Pump characteristics. - The general range of pump characteristics shall be in accordance with table II. Characteristics of pumps to be procured shall be specified in the individual contracts or orders,

Table II - Pump characteristics.

Discharge pressure	Rated capacity	Suction lift at rated capacity[1]	Viscosity range of liquid for test purposes	Range of liquid temperatures under operating conditions[2]	Preferred type of drive
P. s. i. g.	G. p. m.	Inches of mercury	SSU		
50-150	100-1250	5-25	450-7000	35-120	Motor or geared turbine

[1] Referred to centerline of pump rotors.
[2] Temperatives do not necessarily correspond with the viscosity range, but indicate temperative pumps may be expected to encounter in service.

3.5 Pump drivers. -

3. 5.1 Pumps may be direct connected or geared to the driving units as specified or approved in each case by the bureau or agency concerned.

3.5.2 Turbines - Turbine drives shall be in accordance with Specification MIL-T- 17523, and shall have the characteristics as specified in the contract or order.

3.5.3 Motors. - Motor drives shall be of the type and characteristics specified in the contract or order and shall conform to Specification MIL-M-17413 for direct current motors, and to Specification MIL-M-17060 for alternating current motors,

3.5.4 Reduction gears. - Where gears are used, they shall be of the spur, single or double helical, or worm type in accordance with Specification MIL-G-17859 or Specification MIL-G- 18121, as applicable.

3.5.4.1 Reduction gears shall be independent of the pump and connected thereto by a flexible coupling. Such reduction gears may, however, be an integral part of the driving unit.

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3.5.5 Driver rating (motor), - The horsepower rating of each motor shall be not less th that required by the pump under maximum power conditions as determined by actual tests of the identical or similar equipment. If such data is not available the required horsepower rating of the motor shall be determined by the following formula:

$$\text{Horsepower of driver} = \frac{1.05 \text{ (max. HP. output of the pump)}}{\text{(eff. of pump)}}$$

The actual motor rating shall be in accordance with Navy standard motor ratings.

3.5.6 Controllers, - Motor controllers shall be in accordance with Specification MIL-C-2174 for direct current and Specification MIL-C-2212 for alternating current, and shall have the characteristics as specified in the contract or order.

3.5.7 Diesel engine drivers shall be in accordance with Specification MIL-E-15163 and shall have the characteristics as specified in the contract or order,

3.5.8 Gasoline engine drivers shall be as specified in the contract or order,

3.6 Identification plates and information plates. -

3.6.1 Identification plates and information plates shall be furnished on each pump, These plates shall be type A or B in accordance with Specification MIL-I- 15024, except that identification plates of aluminum *or* plastic shall not be used.

3.6.2 Identification plates and information plates shall be firmly secured to the equipment with corrosion resistant metallic screws.

3.6.3 Pump identification plates shall contain data as follows:

(a) Manufacturer's name.
(b) Manufacturer's model or type and size,
(c) Service application,
(d) Manufacturer's serial number,
(e) Salient design characteristics:
    (1) Capacity in gallons per minute.
    (2) Discharge pressure.
    (3) Suction lift,
    (4) Speed of shaft in revolutions per minute.
    (5) Brake horsepower.
    (6) Test pressure,
    (7) Viscosity range (operating),
(f) Contract number (and item number for multiple unit orders),
(g) Bureau of Ships stock number,
(h) Section for inspector' s stamp,

3.6.4 Each driving unit and each accessory unit shall have an identification plate in accordance with the applicable equipment specification,

3.7 Drawings, -

3.7.1 Drawings shall be in accordance with type I of Specification MIL-D-963.

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3.7.2 Outline drawings of the units (pump and driver) shall refurnished on each contractor order; no detail of parts shall be shown on outline drawings.

3.7.3 Pump master drawings shall include detail drawings of all parts, and sectional assembly drawings with lists of material comprising all parts.

3.7.3.1 Detail drawings shall contain a number of details on each sheet; details of a s ngle pump part on size A sheets, resulting in a multiplicity of drawing sheets, shall be avoided.

3.7.3.2 The sectional assembly drawing shall contain a complete list of material with references to detail drawings, The assembly drawing shall not include couplings or brackets which may vary with different contracts or orders.

3.7.3.3 Any subassembly made up ‘of parts which require special alignment or assembly methods which cannot be disassembled, repaired and reassembled onboard ship without the use of special tools and jigs shall be indicated as a subassembly in the list of material.

3.7.4 On detail drawings, surface roughness, waviness and lay shall be specified in accordance with requirements and symbols specified in Standard MIL-STD- 10.

3.7.5 Drawings for drivers and associated equipment shall be in accordance with their respective specifications.

3.8 Technical manuals, -

3.8.1 Technical manuals shall be in accordance with type C of Specification MIL-T- 15071,

3.8.2 Identification piece numbers of parts referred to in technical manuals shall be identical to the identification piece numbers shown on approved master drawings.

3.8.3 Technical manuals shall contain reproductions of dimensioned outline drawings of all units.

3.8.4 Complete characteristic curves based on performance acceptance tests shall be included in each technical manuals.

3.8.5 Preliminary technical manuals submitted for approval shall include all proposed sections of the books,

3.8.6 Two copies of technical manuals shall be packed with each pump unit. Additional copies shall be furnished as specified (see 6, 1). If final technical manuals are not available at the time pumps are shipped, then 2 copies of preliminary technical manuals shall be packed with each pump unit, The final books shall be delivered at a later time as specified in Specification MIL-B- 15071.

3.9 Repair parts and tools. -

3.9.1 Onboard. - Repair parts shall be furnished and processed in accordance with Specifica - tion MIL-R-15137.

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3.9.2 Each set of repair parts and tools specified in the contractor order shall contain the parts listed in table HI, plus any additional parts or tools considered by the bureau or agency concerned, or by the manufacturer, to be essential in successful servicing of the pumps. Unless otherwise specified in the contract or order, the quantity of repair parts per set shall be based on one pump per ship (one pump per set of repair parts).

Table III - Onboard repair parts.

Item	Name of part	Quality per set
a	Rolling contact bearings	100 percent complete replacement for all installed units per vessel
b	Oil seals	
c	Packing, if special, and packing spacers	
d	Coupling bushings, pins, springs, and other wearing parts	
e	Sliding surface bearings	One to four units per vessel, 100 percent replacement for one pump Over four units per vessel, 100 percent replacement for two pumps
f	Cams, slides, slide pins, housings or liners, and similar wearing parts	
g	Timing gears, reduction gears and pinions	100 percent replacement for one pump
h	Rotors, complete, including shaft and coupling complete	

3.9.2.1 Cylinders or bodies shall be supplied, in accordance with item (f) of table III for pumps not equipped with housings or liners,

3.9.2.2 Parts required by item (h) table III are in addition to the parts required by the other items.

3.9.3 Repair parts such as those for turbines, reduction gears, engines, motors, and controllers, shall be furnished in accordance with the applicable component specifications.

3.9.4 Special tools and wrenches, -

3.9.4.1 Each set of onboard repair parts shall include a complete set of all special tools and wrenches required for maintenance of the pumps, gears and drivers,

3.9.4.2 One set of special operating tools and wrenches shall be furnished with each pump unit, and shall include all wrenches and tools required for the operation of the complete pump units. Each set of special operating tools and wrenches shall be furnished in separate steel boxes, If there are no more than two such tools per set they may be attached to the pumps in a secure but readily accessible manner, rather than boxed.

3.9.4.3 Each tool or wrench shall be indelibly marked indicating the intended purpose.

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3.9.4.4 Tools which are available in the Catalog of Naval Material, General Stores Section, such as common wrenches and standard pullers, will not be required. Standard complete or limited sets of pullers or parts of puller sets in accordance with Specification GGG-P-781 shall be identified as to type and use in notes on the outline or section assembly drawings. Special tools shall be detailed, and included in the list of material.

3.9.4.5 Each box containing onboard repair parts shall contain a list entitled "List of onboard repair parts and tools", The list shall be in a format suitable for use on outline drawings and in the instruction books, It shall not be less than nominal 8-1/2 by 11 inches in size.

3.9.4.5.1 The format of the list shall include a heading and columns of data for the items listed, The heading shall include titles and applicable entries as follows:

Reproduced from drawing number _____
Number of ships _____
Application _____
Contract number _____
Manufacturer _____
Quantities are for _____ units per ship
The columns shall include:
Piece number _____
Name of part _____
Quantity _____
Drawing number _____
Standard Navy stock number _____
Additional columns may be used as applicable

3.9.4.5.2 The list shall be so treated as to be resistant to oil, water and fading.

3.9.5 Stock. - The selection, stock numbering and quantities of repair parts for stock shall be determined and processed in accordance with Specification MIL-R-15137.

3.9.5. i Unless otherwise specified in the contract or order, the cognizant supply demand control point for repair parts for stock shall be Ships Parts Control Center, Mechanicsburg, Pennsylvania.

4. QUA LITY ASSURANCE PROVISIONS

4.1 Classification of tests, - Tests shall be classified as follows:

(a) Qualification tests.
(b) Performance acceptance tests,
(c) Hydrostatic tests,

4.2 Qualification tests at a Government laboratory. - Qualification tests shall be conducted at a Government laboratory designated by the Bureau of Ships. These tests shall consist of the tests specified in 4.2.7,

4.2, 1 Pumps submitted for qualification tests shall have characteristic ratings of not less than the following:

Rated capacity	200 g. p. m.
Discharge pressure	100 p, s. i.
Suction lift	15 inches Hg. at 450 SSU.
	25 inches Hg. at 7000 SSU.

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4.2.2 Qualification tests shall be conducted to determine that performance and design of the pumps are in compliance with this specification. The pressure, capacity, and other characteristics of the unit for the qualification test shall be as arranged between the Bureau of Ships and the exhibitor.

4.2.3 To be granted qualification approval as the result of service tests the following requirements shall be met:

(a) Two or more pumps of the same general design and general operating characteristics must have given satisfactory service onboard Naval vessels for a period of at least two years and must still be in service on such vessels at the time qualification approval is requested,

(b) When the design of pump in service, for which the manufacturer request qualification approval, does not meet current Naval requirements as regards suction lift, discharge pressure and capacity characteristics, the manufacturer will be required to furnish satisfactory evidence that he has produced pumps of a design, suction lift, discharge pressure and capacity characteristics equal to those currently required for Naval service. The Bureau of Ships may require additional qualification approval tests of any design of pump which in its opinion has been changed sufficiently to require further demonstration of its suitability.

(c) Any manufacturer whose product has been placed on the qualified products list as a result of service test will be required, on his first contract or order, after being granted qualification approval to submit a complete unit to a Government laboratory designated by the Bureau of Ships for a check test which will consist of the tests conducted for qualification,

4.2.4 Separate qualification tests will be required on vertical and on horizontal units of each type, except that qualification granted on vertical design may be extended also to cover horizontal design at the discretion of the Bureau of Ships.

4.2.5 Units submitted for qualification test shall be operated at the maximum rubbing speed for which the unit is designed. Approval of a design of pump operating at a rubbing speed greater than that at which the tested unit operated on qualification approval test will not be given without an additional life test (see 3.3.16 and 4.2.7. 1), The life test specified in 4.2, 7, 1 is intended to demonstrate the reliability of the type.

4.2.6 Unless otherwise arranged between the Government and the exhibitor, the pump for the qualification test shall be furnished complete with a prime mover and necessary accessories. The prime mover need not be made to Government specifications; however, see 4, 3, 10.3, Qualification of a pump does not include qualification of the driver,

4.2.7 The following tests, in the order listed shall be made to determine qualification:

(a) Performance tests to demonstrate the ability of the pump to operate at the design characteristics (see 4.2. 2), Sufficient data shall be taken during these tests to prepare pump characteristic curves as specified in 4.3.5, except that curves of capacity versus power input to the driving unit, as specified in 435 (e), will not be required,

(b) Additional operating runs under c conditions to be selected by the Bureau of Ships to determine:

(1) Volumetric efficiency, (2) mechanical efficiency and (3) capacity, at varying speed,

(c) Dismantle pump and measure all parts subject to wear and erosion.

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(d) Life test (see 4.2.7.1).
(e) Dismantle pump to measure extent of wear and erosion and to note general condition of pump,
(f) Duplicate runs under (a) and (b) to determine loss of volumetric efficiency, mechanical efficiency and capacity due to wear (see 4.2.7.2),
(g) Shock test (see 4.3.10.1).

4. 2.7.1 The life test shall consist of a run of 50 hours' continuous duration using 2 mixture of 1 percent fuel oil and 99 percent fresh water, temperature about 100" F., at rated discharge pressure of the pump and with a suction lift of 5 inches mercury, at the maximum rated speed of pump under test (see 3.3.8.9.1.1),

4.2.7.2 A comparison of the mechanical and volumetric efficiencies and the capacity of the pump before and after the life test shall be made to determine loss in efficiencies and capacity due to the life test. If such comparison shows that the efficiencies or capacity of the pump have not been excessively reduced in comparison, with other pumps of the type designed for the same service, and if the wear on any parts indicates that it will not result in complete failure of the pump, the unit will be considered suitable for use in Government service provided all other requirements of this specification are complied with. If, however, during the test, complete failure occurs for any reason, the unit will be rejected.

4.3 Performance acceptance tests. -

4.3.1 The driving unit shall be tested as required by the applicable specifications for turbines, motors, and gasoline or Diesel engines. During these tests the pump shall be operated at sufficient load to demonstrate conformance to the suction lift and discharge pressure requirements.

4.3.2 AU pumps, except those driven by alternating current motors, shall be tested at 25 percent over normal speed by a continuous nonstop run of at least 30 minutes; while running at this speed the pump need not handle load as this is to insure proper alignment and balance. These tests may be conducted without the' driving units, providing the driving units, including reduction gears, if used, have been independently tested in the same manner.

4.3.3 The rotors, shafts and timing gears, including repair parts of all pumps, shall be tested for smoothness of running at all speeds, including the 25 percent overspeed test of at least thirty minutes.

4.3.4 One pump of each size and design as selected by the Government inspector, complete, including the turbine or the motor and controller, and reduction gear, if any, shall be given a test to determine the overall power or steam consumption of the unit under the conditions of the contract, This test may be conducted at the shop of the main contractor or subcontractor as arranged with the bureau or agency concerned, The test ordinarily shall be conducted at the same time as, and shall include, the inclined operation requirements as specified for the driving units; (see Specifications MIL-T-17523, MIL-M-17413, MIL-M-17060, MIL-G-17859, MIL-G-18121 and MIL-E-15163). For inclined operation tests, the unit need not be fully loaded, but shall be run at maximum operating speed and loaded sufficiently to prevent damage to the unit, The functioning of the speed governors of turbine driven pumps shall be carefully observed during these tests, and all adjustments that may be necessary to insure operation satisfactory to the bureau or agency concerned at all normal speeds shall be made before the units are accepted, The turbine governors shall function as specified in Specification MIL- T- 17523. When pump pressure regulating governors are specified, (see 6. 1) the foregoing test procedure shall apply also to this equipment. In this test the ability of the pump to handle its rated capacity of specified liquid at the maximum and minimum viscosities and maximum suction lift specified shall be adequately demonstrated.

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4.3.5 Sufficient data shall be taken during the tests of 4,3.1, 4,3.2, 4,3.3 and 4,3,4 to prepare pump characteristic curves as specified below. The curves shall be corrected. in all cases to the steam or the electric power operating conditions specified in the contract or order. Complete operating characteristics of the pump and its driver, as specified in the contract or order, as well as the operating characteristics of the unit during the particular test run, shall be shown on each curve sheet. The capacity shall be expressed in gallons per minute, the discharge pressure in pounds per square inch gage, the suction lift in inches of mercury, the viscosity in Saybolt seconds universal (unless otherwise specified in the contract or order) and the speed in revolutions per minute, Curves shall be furnished as follows:

(a) At least two curves of capacity versus pump speed at constant specified discharge pressure and at constant specified maximum suction lift when handling specified fuels as follows:
  (1) One curve at maximum specified viscosity,
  (2) One curve at minimum specified viscosity. In the event that the suction lift at the minimum specified viscosity is not specified in the contract or order and in the event the minimum specified viscosity is not less than 100 Saybolt seconds universal, then the suction lift at the minimum viscosity shall be 80 percent of the specified suction lift at the maximum specified viscosity.
  (3) One curve at each intermediate viscosity that may be specified in the contract or order.

(b) A curve of capacity versus pump mechanical efficiency for each of the curves obtained under item (a),

(c) A curve of capacity versus volumetric efficiency for each of the curves obtained under item (a),

(d) A curve of capacity versus horsepower input to the pump for each of the curves obtained under item (a),

(e) A curve of capacity versus power input to the driving unit, expressed in pounds of steam per hour or electrical horsepower as applicable, for each of the curves obtained under item (a).

(f) One curve of capacity versus suction lift at specified maximum discharge pressure and at rated speed for each of the viscosity values specified in item (a),

(g) In the event that two or more rated capacities are specified, a set of curves, which are required by above subparagraphs (a) to (f) inclusive, shall be furnished for each specified rated capacity.

4.3.6 An exception to 4. 3.5 will be permitted in the case of pumps driven by single speed alternating current motors in which case the pump may be run at constant speed and a curve of capacity versus pump discharge pressure submitted in lieu of the curves required under item (a). The curves of efficiencies and power shall be plotted versus discharge pressure in lieu of versus capacity as required under items (b), (c), (d), and (e).

4.3.7 In order to simplify the foregoing test procedure, the Bureau of Ships when requested, may authorize separate tests of the driving unit (including reduction gear, if any), and pump at the works of different manufacturers and will accept overall power or steam-consumption results and characteristic curves computed from the results thereof, However, this shall not be construed in any way to release the primary contractor from an operating run on one completely assembled unit from each contract or contracts for identical units. This operating test' shall be run with the unit fully loaded, and shall demonstrate compliance with all requirements of this specification as to operation of constant pressure and turbine governors, thrust, balance, lubrication, at maximum rated speed.

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4.3.8 The capacity of the pump (selected as specified in 4.3. 4) may be determined by any convenient method that will insure obtaining results with probable errors not greater than 2 percent, All gages shall be calibrated before and after the tests or oftener, if necessary. The specified viscosities shall be maintained within 5 percent regardless of temperatures,

4.3.9 Reports. -

4.3.9.1 Complete performance acceptance test reports, including copies of recorded test data and characteristic curves plotted therefrom and covering the following shall be furnished as specified in 4.3.9.2:

(a) Test report of driving unit.
(b) Test report of controller (if involved).
(c) Test report of pump.
(d) Test report on the combined assembled unit, including a definite report on the suitability of the driving unit (and controller or pressure regulating governor, if furnished) for its specific application and harmony of assembly of all independent units going to make up the complete equipment,

4.3.9.2 The reports specified in 4. 3.9.1 shall be distributed as follows:

(a) Three to the Bureau of Ships (to be forwarded via the cognizant inspector).
(b) One for primary district inspection office.
(c) One for branch inspection office (if any).
(d) Two to each supervisor of shipbuilding concerned (one for supervisor's files, the other for the shipbuilder).
(e) One to the Commander of each ship building yard concerned; for Navy yard built ships, when specified in the contract or order,

4.3.10 Shock tests. -

4.3.10.1 One complete pumping unit of each type, design and size shall be subject to the high impact shock test of Specification MIL-S-901 as specified in 3, 3.2.

4.3.10. 2 Shock tests will be conducted at the manufacturer's plant. Where a manufacturer is unable to conduct these tests at his own plant, he may arrange to have them conducted at a commercial laboratory, or Government laboratory suitably equipped to conduct tests, Where shock tests are conducted at a Government laboratory, copies of the applicable master drawings shall accompany the units.

4.3.10.3 All pump units shall be shock tested with drivers unless otherwise approved by the bureau or agency concerned. Flexible coupled pumps shock tested with one driver will not be required to be shock tested again when supplied with a different driver. Prime movers are subject to shock tests in accordance with the applicable equipment specifications.

4.3, 10.4 Compliance with shock resistance requirements may be covered by test of a similar unit subject to bureau or agency concerned approval,

4.3.10.5 Equipment previously accepted will not be resubmitted for shock test except when evidence of low shock resistance develops in the units installed, The bureau or agency concerned will initiate action with the manufacturer for correction of deficiencies, and may require shock tests at any time,

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4.3.10.6 The pump unit shall be considered to have failed to pass the shock tests in the event of any of the following:

(a) Breakage of any parts, including mounting bolts.
(b) Appreciable distortion or dislocation of any part, such as shaft, mounting feet, bearings,
(c) A mechanical unbalance of more than two times the amplitude of unbalance measured prior to tests at rated speed, or more than that permitted by the specifications for the driver,

4.3.10.6.1 Pump units which have beep subjected to the high-impact shock test and have failed to conform to the requirements of this specification will not be acceptable.

4.3.10.7 Pump units which have been subjected to the high-impact shock test and have successfully passed this test shall be considered acceptable for such service as the bureau or agency concerned may authorize. Such equipment shall be reconditioned by the manufacturer as follows:

(a) Minor deformations of mounting flanges shall be corrected,
(b) Minor deformations affecting alignment shall be corrected,
(c) All bearings shall be replaced,
(d) Each part shall be carefully examined by the contractor and any part which he considers substandard shall be replaced.

Minor deformations shall be defined as those which do not cause unqualified rejection of the design under high-impact shock test but which are in excess of the design dimensional tolerances specifies on the applicable pump drawings.

4.4 Hydrostatic tests. - Each pump casing shall be tested hydrostatically to one and one-half times the maximum rated discharge pressure, but in no case less than 50 p.s, i, A light grade of mineral oil, not exceeding 200 second Saybolt universal (at 130° F.) may be used for hydrostatic tests,

4.5 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of MateriaL

5. PREPARATION FOR DE LIVERY

5.1 Preservation, packaging, packing and marking shall be in accordance with Specification MIL-P- 16789 and as specified (see 6, 1),

5.2 Onboard repair parts and tools shall be furnished in repair part boxes, if boxes are required conforming to type M or type W of Specification MIL-B-233 (see 6. 1).

6. NOTES

6.1 Ordering data, - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Installation:
  (1) Horizontal or vertical mounting,
  (2) If vertical, whether case mounted or table mounted,
  (3) Size of the inlet and outlet connections or a statement that these shall be as approved by the bureau or agency concerned,
  (4) Limits of length, width, and height for disassembly of unit for normal maintenance overhaul, if limits are required for particular installation.
  (5) A statement as to the location of inlet and outlet connections.

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(c) Pump characteristics:
  (1) Rated capacity,
  (2) Discharge pressure.
  (3) Suction lift.
  (4) Characteristics of liquid:
    (a) Viscosities,
    (b) Liquid temperatures.
(d) Type drive.
(e) Turbine characteristics (for turbine drive):
  (1) Type of connection to driven unit.
  (2) Steam conditions (pressure and temperature) at the throttle and the maximum designed boiler steam drum pressure of the installation for which the turbine is intended.
  (3) Exhaust conditions (pressure or vacuum) at turbine exhaust.
(f) Electrical characteristics (for motor drive):
  (1) Voltage, phase characteristics and frequency,
  (2) Service class.
  (3) Ambient temperature,
  (4) Duty,
  (5) Enclosure.
  (6) Winding (d. c.) or design of motors (a. c.).
  (7) Bearings
  (8) Insulation.
(g) Internal combustion engine characteristics (for engine drive).
(h) A complete list of all items of equipment, not essentially parts of the pump, which the pump contractor is required to furnish. Usual mention should be made in such a list of repair parts and tools. The following items should be considered also if it is required to have the pump manufacturer or contractor supply them:
  (1) Relief valves or relief by-pass valve.
  (2) Drain valves.
  (3) Suction and discharge valves.
  (4) Pressure vacuum or combined gages.
  (5) Thermometers and thermometer sockets,
  (6) Permanently connected revolution indicators,
  (7) Air or vent valves and cocks,
  (8) Motor controllers.
  (9) Pressure regulating governors,
  (10) Technical publications required,
(i) Type of preservation and packing required and type of onboard repair parts boxes (type M or type W of Specification MIL-B-233).
(j) Data required with bids.

6.2 In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Qualified Products List, The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification, Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Ships, Navy Department, Washington 25, D. C.

6.3 This specification supersedes that part of Section S47-3 of the General Specifications for Machinery covering pumps for class N-1 service.

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Patent notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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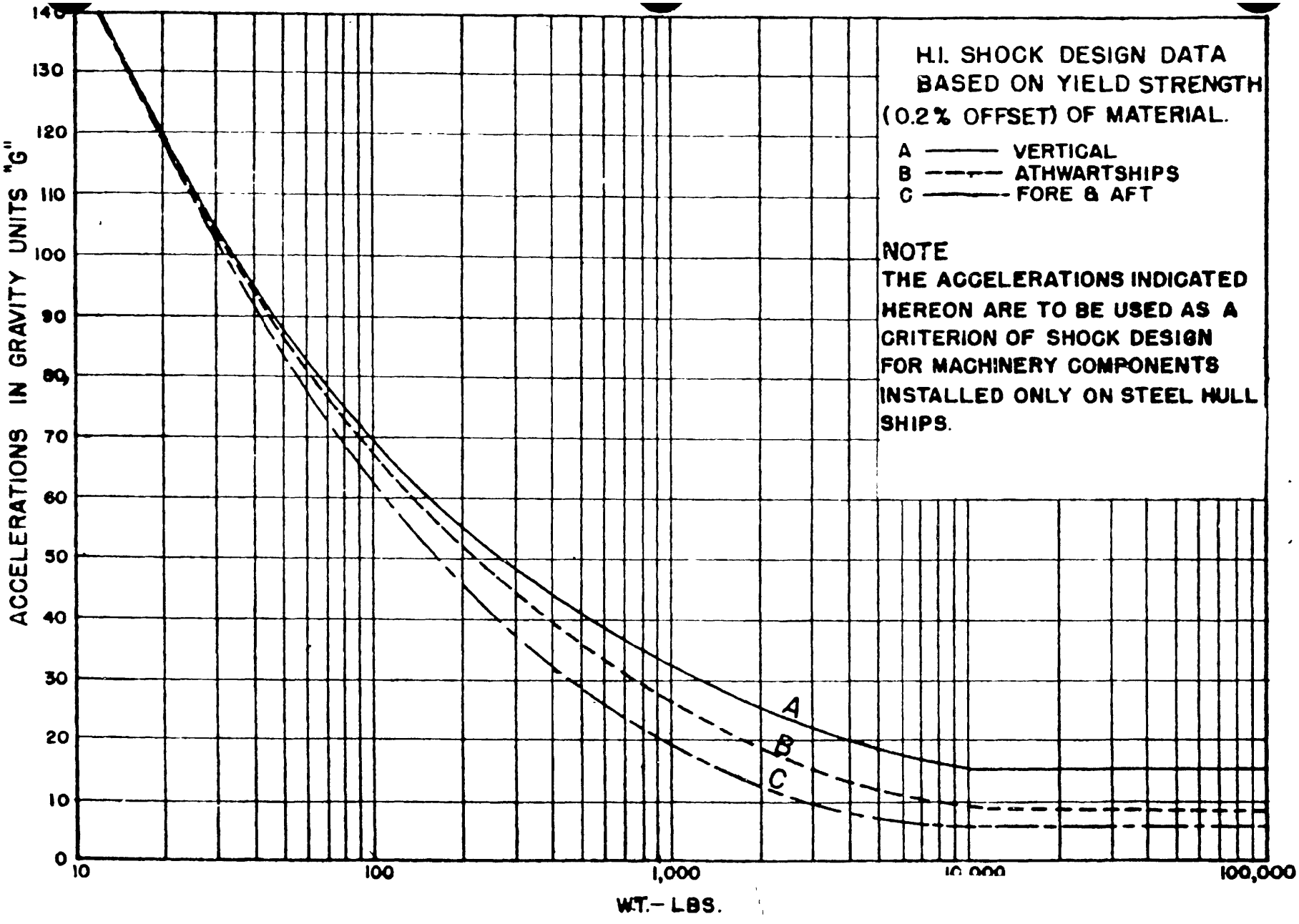


Figure 1. - H. I. shock design data.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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